

## im 2 lec 11 (10 apr)

5C of marketing

Company, Customer, Competition, Collaborators, Context

- customer - demography, life-style, age
- positioning statement - write the single most important thing for each of the criteria
  - for (target audience)
  - our product/brand (competitive fame/category)
  - that provides (single most important benefit)
  - because (reason to believe)
- brands - name/term/design/symbol intended to identify the goods and services of one seller from the competitors
- subtly, subconscious, long term, promise
- brand identity prism (physique, persona, relationship, culture, user image, self-image)

# im 2 lec 1 (16 , 23 jan)

## contents

structure of an organization

types of organization

---

- Organization - group of individuals working together for one objective on a continual basis
- culture - set of unwritten and unsaid rules
- grapevine - encourage to get real criticism from the people in an organization

## mechanistic

- organic structure
  - defines the position and function
  - what may work in one demographic may not work in another
  - defined role for each person
- something that may work in a smaller organisation wont work in a larger organisation
- flat organisations - IT firms
- rigid structures - for standardised procedures
- commissioning a plant - team based
- combination of different types of structures for the maximum efficient

## greiners model

assumptions

1. every company grows (growth)
  1. companies need to change in order to grow
  2. some are reluctant to do so
2. management style
  1. clan style
  2. competitive style
  3. hierarchical style
3. structure
4. coordination

## phases of growth

1. growth through creativity - leadership crisis
2. direction - autonomy
3. delegation - control
4. coordination - red tape crisis - red tape is essentially a perspective of A department about some other B department which may not always be positive
  1. staff function - advisory not actual work
  2. line roles - functional department
5. collaboration - internal growth
6. alliances - identity

## **question**

how can u apply greiner's model to xyz company? (google, facebook etc)  
(dont describe greiner's model)

## **Organizational design**

- create value for the future
- people at the centre
- design is like a map, implet gps
- build resilience for unknown
- no perfect model

## **parts of an organizational design**

1. mission
2. process
3. roles
4. structure
5. people and practices

## **finance**

- dont print more money
- deficit budget
- global perspective

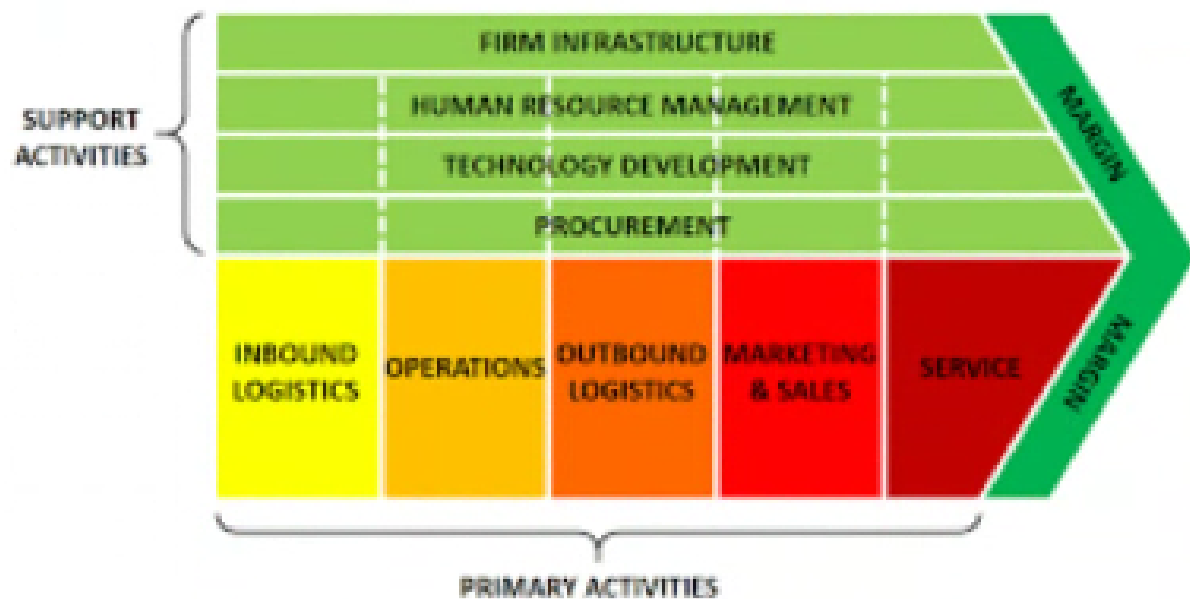
## im 2 lec 3,4 (6 feb, 13 feb)

- retained earnings - paying yourself
- tangible vs intangible assets
- assets vs liabilities
- intangible assets - income tax
- good in process
- contingent liabilities
- depreciation and amortization in profit and loss not in balance sheet
- for which sector/industry -> which factors are important

## porter's model

1. number of competitors
2. industry growth
3. similarities in products/services offered
4. exit barriers
5. fixed costs

## value chain



—C

## differentiation

- two things to strive for - quality and price
-

# im 2 lec 5 (27 feb)

## Contents

---

## marketing mix

1. promotion
2. product
3. place
4. price

## im 2 lec 7 (16 march)

lec 6 - missed

### MOC for some fluids

#### sulphuric acid

Moc for reactors and pipelines

| fluid     | conc    | temperature              | MOC                             |
|-----------|---------|--------------------------|---------------------------------|
| $H_2SO_4$ | 80-98%  | $<60\text{ }^{\circ}C$   | Mild steel                      |
|           | $<80\%$ | $60-80\text{ }^{\circ}C$ | SS                              |
|           | any %   | $<40\text{ }^{\circ}C$   | Rubber lined, Polypropene, HDPE |
|           | $<40\%$ | high temperatures        | Glass lined                     |

- pumps - cast iron or alloy 20 (much cheaper, can be definitely used for pumps)
- for high temperature - Hastalloy pumps
- storage - MS

#### nitric acid

| fluid   | conc | temperature                     | MOC                 |
|---------|------|---------------------------------|---------------------|
| $HNO_3$ | 98%  | upto $150-160\text{ }^{\circ}C$ | SS304 (reactor)     |
|         |      |                                 | Aluminium (storage) |
|         | any  | any                             | SS304 (pumps)       |
|         |      |                                 |                     |
|         |      |                                 |                     |

- chloro compound and nitric acid -> don't use SS
- for heat exchangers - tube support and tubes are high quality SS
- outside body is mild steel
- jacket for a reactor is not SS
- MS limpet, particular type of welding rod

#### hydrochloric acid and caustic soda

| fluid      | conc                                                                  | temperature                                         | MOC                                                                         |
|------------|-----------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------|
| <i>HC</i>  | 30 % (more than 30% fumes)                                            | moderate temperatures<br>PP - 70 °C<br>HPDE - 40 °C | rubber lined tanks and pumps,<br>PP pipe lines,<br>HPDE pipelines,<br>drums |
|            |                                                                       | high temperatures for heating HCl                   | glass lined, hastalloy                                                      |
|            |                                                                       | cooling HCl                                         | direct ice, or cooling coils inside the reactor                             |
| <i>NOH</i> | 48% (highest with reasonable viscosity, crystallisation at low temps) | 120 °C                                              | MS                                                                          |
|            |                                                                       | above 120                                           | MoNel, Ni                                                                   |
|            |                                                                       |                                                     | <b>cannot use glass or glass lines</b>                                      |

- Valves and pumps - cast iron, or with teflon ceiling
- no heat transfer + 120 deg C - PTFE

## ammonia

- cylinders - CI or MS stored as a liquid
- pipes - two lined (liquid phase)
- ammonia reactions - Cu catalyst - SS
- ammonia - 28% handling without pressure

## organics

- oxidation - Co, Br, Mg - HBr formation, so Titanium is used for reactors, pumps either Ti or Hastalloy. Ti can be welded asw, initial High project costs
- acetic acid, acetaldehyde - no chloride ions, SS otherwise same as HCl
- unknown - SS304

## pumps

- what kind of pump?
- centrifugal - high flow rate
- gear - low flow rate
- vacuum pump - types



- oil pump - 0.1-1 mm (high vacuum but poor capacity)
- water ring - barometric leg , 660 mm
- ejector (merchant)

## motor

- 3000 rpm standard
- gear box - reduce rpm without reducing torque

## utility pipe

- cw pipeline, top floor higher diameter

## effluent treatment

- have an energy balance
- water balance, water meters
- evaporation losses
- acid and base balance, acid at one collection tank, base at another. **Neutralise**
- estimate COD - do the simple thing first, aka coagulation using alum
- don't go for chemical treatment, if pH is right go for biological reactor (48-72 hours)
  - bio reactors - control point is how much bacteria is alive
- waste solids - incineration
- evaporator - MEE, heat pumps
- dilution - when u have steam condensate it can be used as water for toilets or gardens
- boilers - more smaller boiler > one big boiler
- boiler feed water is measured for steam consumption measurement

## **im 2 lec 9 (27 mar)**

- quality assurance and audits
- processes involved (purchase, stores, operations, QC)
- inputs
  - people
  - procedure
  - infrastructure
  - materials/reagents
  - calibrations (for instruments)
  - measure responses
- output of QC - report, consistent analysis